

## TECHNICAL DOCUMENTATION — CerebroStim X1

Device: CerebroStim X1 | Class IIb active implantable neural stimulator | Non-IVD | Active | Software present

### SECTION 1 — SAFETY AND GENERAL PRINCIPLES

The CerebroStim X1 is an implantable neurostimulator for management of treatment-resistant epilepsy. Risk management per ISO 14971:2019 covers all phases of device lifecycle.

Benefit-risk conclusion: clinical benefits of seizure reduction outweigh risks of surgical implantation and device malfunction for eligible patients.

### SECTION 2 — BIOCOMPATIBILITY (EP 7 — PARTIAL)

Materials have been characterised per ISO 10993-1:2018.

7.1 Chemical composition: titanium housing, platinum-iridium electrodes, medical grade epoxy encapsulant.

7.2 No materials of biological origin are used.

7.3 Cytotoxicity and genotoxicity testing: negative for ISO 10993-5 and ISO 10993-3.

7.4 Device is not flammable.

7.5 Materials in contact with tissue: titanium and platinum-iridium have established biocompatibility profiles in chronic implant use.

NOTE: Leachable and particulate contamination assessments (7.6, 7.7) are in progress and are not available in this version of the technical documentation.

### SECTION 3 — ELECTRICAL SAFETY AND SOFTWARE (EP 12 — PARTIAL)

IEC 60601-1:2018 testing complete for electrical safety (12.1).

Mechanical safety testing per IEC 60601-1 complete (12.2).

EMC testing per IEC 60601-1-2 complete (12.13).

Software: IEC 62304 Class C software lifecycle documentation complete (12.9).

Software version V1.2.0 is indicated on device label and patient documentation (EP 13B satisfied).

NOTE: Cybersecurity assessment (12.8) was initiated but penetration testing is not yet complete. Cybersecurity documentation is in draft.

NOTE: Requirements for 12.3-12.7, 12.10-12.12 testing are in progress.

### SECTION 4 — PATIENT IMPLANT CARD (EP 13A)

A patient implant card is provided with each device in accordance with Schedule 1 clause 13A. The card includes: device name, model number, serial number, implant date field, surgeon field, implant site, and MRI conditional status.

### SECTION 5 — INFORMATION LABELLING (EP 13)

Labelling complies with Schedule 1 clause 13 requirements including device identification, manufacturer details, sterilisation indicator, single-use designation, and instructions.

## SECTION 6 — CLINICAL EVIDENCE (EP 14)

Clinical evidence strategy: direct clinical data from investigational device study.

Literature review (MEDLINE, EMBASE, Cochrane, 2000-2025): 24 relevant studies identified.

Clinical investigation: CEREBROSTIM-AU-001 study (n=45, 12-month follow-up) demonstrates mean seizure reduction of 58% vs baseline ( $p<0.001$ ). No serious device-related adverse events in the study cohort.

Post-market clinical follow-up plan: Annual registry data collection from implanting centres. PMCF protocol approved by ethics committee.

## SECTION 7 — UDI AND TRACEABILITY

UDI assignment: in progress. CerebroStim X1 is a Class IIb device; UDI compliance required by 1 July 2027. UDI system selection completed (GS1 GTIN). Submission to GUDID pending. Current documentation does not yet include UDI-DI or UDI-PI.